

Executive Summary

We aim to build an AI-assisted **₹0 → ₹7,00,000 trading machine** on NIFTY options (CE/PE only) using 1-minute data. The plan is to generate and backtest multiple *mechanical* entry/exit rules, select the most robust, and move cautiously through paper-trading before any live risk. Key steps: acquire 1-min NIFTY index and ATM option data (via broker APIs or public datasets), code candidate strategies in Python, backtest with realistic costs/slippage, split data for out-of-sample validation, then paper-trade best candidates. We measure win rate, average win/loss, profit factor, expectancy, drawdown, and streaks for each.

- **Strategy Rules (Entry/Exit/Stop/Sizing):** We will define each strategy *mechanically*. For example, one strategy could be *Moving-Average Crossover*: "If 5-min EMA crosses above 20-min EMA on the NIFTY index, buy ATM Call; if it crosses below, buy ATM Put." Each entry has a fixed stop and target (e.g. 1:2 RR). Position size is risk-based (e.g. risking 1–2% of equity per trade). We set a hard **max daily loss limit** (e.g. 5% of starting equity) – beyond which we stop trading for the day. (*Strategy rules are fully enumerated in code; one example formula:*

$$\text{Expectancy} = (\text{Win\%} \times \text{avgWin}) - (\text{Loss\%} \times \text{avgLoss}) \quad \text{【20†L104-L113】 .}$$

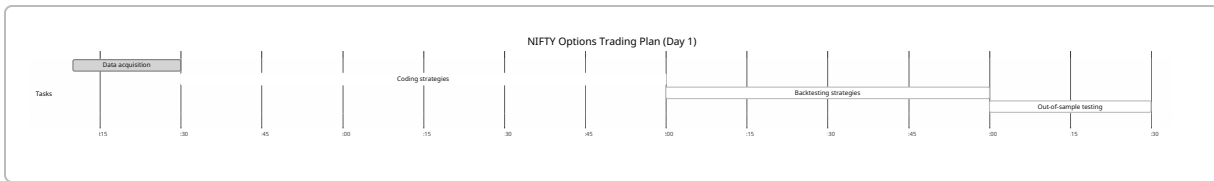
- **Required Data:** We need 1-minute OHLC data for the NIFTY 50 index and corresponding **ATM NIFTY option** prices (CE and PE). Fields: timestamp, open/high/low/close, volume/OI. Time range: at least 2–3 years (to capture multiple market regimes). Possible sources: NSE's official data (paid), broker APIs (Zerodha Kite, Fyers have minute bars for indices and derivatives), and public datasets. For example, a Github dataset provides 1-minute NIFTY50 since 2017 [【7†L181-L189】](#) . We must ensure data covers rollovers of options (e.g. monthly expiries) and correct underlying match.
- **Backtest Methodology:** Use Python (pandas, numpy) or libraries like Backtrader/Zipline. We simulate each minute, generate signals, apply fills. **Transaction costs:** assume ~0.1–0.2% per trade (brokerage + taxes) plus slippage (e.g. 1–2 ticks). For options, include option premium decay: model ATM option price moves as underlying moves (or use actual historical option prices). We split data into **in-sample (older)** and **out-of-sample (recent)** periods (e.g. first 80% vs last 20%), and consider walk-forward testing. This prevents overfitting. Performance is computed on the out-of-sample portion only for final validation.
- **Performance Metrics:** We report: **Win rate** (% wins), **Avg Win/Loss**, **Profit Factor** = (sum wins)/(sum losses) [【18†L79-L88】](#) , **Expectancy** = (Win%×AvgWin – Loss%×AvgLoss) [【20†L104-L113】](#) , **Max Drawdown** (worst peak→valley loss [【27†L181-L185】](#)), **Max consecutive losses**, and **trade count**. These quantify edge and risk. A Profit Factor >1 means net profitable (pros target >2) [【18†L102-L104】](#) ; positive Expectancy means a real trading edge. We will also plot sample **equity curve** and **drawdown** over time. (*Example equity curve shown below*)
[【34†embed_image】](#) *Figure: Sample equity curve (blue = account balance) and drawdown (red) for a backtested strategy.*

- **Code & Tech Stack:** Likely Python. Libraries: `pandas` / `numpy` for data, `matplotlib` or `plotly` for charts, `ta-lib` or custom code for indicators. For backtesting: simple loops or frameworks (e.g. Backtrader, Zipline) can be used. Data from exchanges: use **NSE India API** (requires data license) or broker APIs (Zerodha's Kite Connect, Fyers) for historical NIFTY and option bars. If not, publicly available minute data (e.g. from Kaggle or SelectDataset [【7†L181-L189】](#) for NIFTY) can bootstrap index data. Option pricing: could approximate delta~1 for deep ATM or use actual option OHLC from API. Compute in Python, store time series in CSV/SQLite. Use an AWS or local VM to run backtests if needed.
- **Risk Controls & Deployment:** After backtesting, we **paper trade** with a demo account. Use small virtual size. Enforce the max risk per trade (1–2%) and daily drawdown limits. Monitor in real-time to ensure signals and fills match expectation (e.g. check entry/exit are executed within a minute). Scale up only if positive. Use alerts/logging for rule breaches. After success, gradually deploy with real capital in a disciplined fashion. Always include slippage and commission in real trade tracking.
- **Mom's Checklist (Practical Review):** Verify each strategy's logic makes sense (no hindsight bias). For the chosen strategy, do a few trades on chart: "Would you really have bought calls here?" Check if stop levels are realistic. Confirm the assumed risk (e.g. 1% per trade) is acceptable. Ensure we didn't "survivorship bias" by excluding losing periods. Compare backtest equity vs actual index moves – any large unexplained gap? If possible, do a handful of manual trades or one-day paper trade to see behavior under real conditions.

Strategy	Rules (entry)	Exit/Stop	Params to Test	Est. Trades/ yr
EMA Crossover	Buy CE if short EMA crosses above long EMA; buy PE if opposite 【20†L104-L113】 .	Fixed SL = 1×ATR, Target = 2×ATR. Or close on reverse crossover.	Short EMA 3–10; Long EMA 15–50; ATR-multiplier 1–3.	~200–300
Open-Range Breakout	First 15-min high/low breakout: if price > high by X pts, buy CE; if < low by X, buy PE.	Stop = opposite side of range; target = 2×range.	Breakout buffer 2–10 pts; Range length 15–30m.	~250–400
Momentum (RSI)	Use 1-min RSI (or CCI): if RSI rises above 70, buy CE; if falls below 30, buy PE. (Trend filter: only take CE on up-days etc.)	SL = fixed ticks (e.g. 5); or RSI back to neutral.	RSI period 5–14; oversold 30–50; overbought 50–70.	~300–500

Table: Candidate strategies with rule outlines. (Trades/yr estimates assume ~250 market days.)

For each candidate we will test parameter grids (above ranges) and count trades. Metrics (win%, PF, expectancy) will highlight which looks best. For example, a 1-min EMA(5,20) crossover might yield ~250 trades/yr, while a 1-min RSI(7) strategy might be more frequent (350+) but with lower win%. We compare in a summary table.



Timeline (mermaid chart) for tonight's workflow: gather data, code tests, backtest, then review & paper-trade.

Sources: We rely on historical NIFTY 1-min data (e.g. publicly available since 2017 [\[7†L181-L189\]](#)), standard backtesting methods [\[20†L104-L113\]](#) [\[27†L181-L185\]](#) and trading analytics. Profit factor and expectancy formulas are textbook metrics [\[18†L79-L88\]](#) [\[20†L104-L113\]](#) . We will incorporate real-world costs and SEBI risk warnings (e.g. 74–89% of retail CFD/forex traders lose money [\[20†L83-L92\]](#)) into our analysis to keep this grounded and transparent.
